

$$\text{Resolver: } y = \left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + \frac{x^2}{2}$$

$$\frac{dy}{dx} = p$$

$$y = p^2 - xp + \frac{x^2}{2}$$

$$\frac{dy}{dx} = \frac{2p dp}{dx} - x \frac{dp}{dx} - p + x$$

$$p = 2p \frac{dp}{dx} - x \frac{dp}{dx} - p + x$$

$$2p - x = (2p - x) \frac{dp}{dx}$$

$$\frac{dp}{dx} = 1$$

$$dp = dx$$

$$p = x + c$$

$$1) y - x \frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2$$

$$\frac{dy}{dx} = p$$

$$y = p^2 + xp$$

$$\frac{dy}{dx} = 2p \frac{dp}{dx} + x \frac{dp}{dx} + p$$

$$x = 2p \frac{dp}{dx} + x \frac{dp}{dx} + p$$

$$0 = \frac{dp}{dx} (2p + x)$$

$$dp = 0$$

$$p = c$$

$$y = c^2 + xc$$

$$2) y + \left(\frac{dy}{dx}\right)^2 = x \frac{dy}{dx}$$

$$\frac{dy}{dx} = p$$

$$y + p^2 = xp$$

$$y = xp - p^2$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p - 2p \frac{dp}{dx}$$

$$p = x \frac{dp}{dx} + p - 2p \frac{dp}{dx}$$

$$0 = (x - 2p) \frac{dp}{dx}$$

$$p = c$$

$$y = xc - c^2$$

$$3) y = x \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\frac{dy}{dx} = p$$

$$y = xp + \sqrt{1+p^2}$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \frac{1}{2}(1+p^2)^{-1/2} p \frac{dp}{dx}$$

$$x \frac{dp}{dx} + p + (1+p^2)^{-1/2} p \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x + (1+p^2)^{-1/2} p)$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc + \sqrt{1+c^2}$$

$$4) y = x \frac{dy}{dx} + \frac{dy}{dx}$$

$$\frac{dy}{dx} = p$$

$$y = xp + p$$

$$\frac{dy}{dx} = \frac{dp}{dx} x + p + \frac{dp}{dx}$$

$$p = \frac{dp}{dx} x + p + \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x+1)$$

$$p = c$$

$$y = xc + c$$

$$5) y = x \frac{dy}{dx} + \sin\left(\frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = p$$

$$y = xp + \sin p$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p + \cos p \frac{dp}{dx}$$

$$p = x \frac{dp}{dx} + p + \cos p \frac{dp}{dx}$$

$$0 = (x + \cos p) \frac{dp}{dx}$$

$$p = c$$

$$y = xc + \sin c$$